

Meta-Regression

This vignette introduces the regression functionality of the `dtametaTMB` package for meta-analysis of diagnostic test accuracy (DTA) studies.

Validation

The subgroup HSROC and Reitsma implementations reproduce the Cochrane Handbook RF and anticcp examples closely, including the baseline parameters (HSROC: accuracy, threshold, shape // Reitsma: logit sensitivity and specificity), between-study variance estimates, and subgroup effects on accuracy and threshold. Small numerical differences may arise from optimizer settings, numerical tolerances, and implementation details, but all benchmark estimates and standard errors were recovered to close agreement.

Dummy-coding vs. cell-means parameterization

For subgroup analyses, we distinguish between two equivalent parameterizations of the same model. In the reference-group parameterization, one subgroup serves as the baseline and the remaining subgroup effects are expressed as deviations from this reference. In the group-specific (cell-means) parameterization, each subgroup has its own parameter directly. The former is convenient for testing subgroup differences, whereas the latter is convenient for reporting subgroup-specific estimates and plotting subgroup-specific HSROC curves. The Reitsma subgroup model is fitted twice with both parameterizations. The subgroup HSROC model is estimated using a reference-group parameterization, and subgroup-specific parameters are recovered by evaluating the fitted linear predictors at subgroup-specific design points via Z_{pred} .

How do we include covariates in the Reitsma model?

For sensitivity in study i , we have the number of diseased individuals testing positive: $y_{Ai} \sim \mathcal{B}(n_{Ai}, \pi_{Ai})$.

Similarly for specificity, we have the number of non-diseased individuals testing negative: $y_{Bi} \sim \mathcal{B}(n_{Bi}, \pi_{Bi})$.

Now we introduce a p -dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\begin{pmatrix} z_i^\top \mu_{Ai} \\ z_i^\top \mu_{Bi} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} z_i^\top \mu_A \\ z_i^\top \mu_B \end{pmatrix}, \Sigma \right), \quad \text{with} \quad \Sigma = \begin{pmatrix} \sigma_A^2 & \sigma_{AB} \\ \sigma_{AB} & \sigma_B^2 \end{pmatrix}.$$

Let's assume that z_i includes a single binary covariate representing two subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$z_i^\top \mu_{Ai} = \begin{cases} \mu_{Ai} & \text{for subgroup 1,} \\ \mu_{Ai} + \nu_{A2} & \text{for subgroup 2,} \end{cases} \quad z_i^\top \mu_{Bi} = \begin{cases} \mu_{Bi} & \text{for subgroup 1,} \\ \mu_{Bi} + \nu_{B2} & \text{for subgroup 2,} \end{cases}$$

$$z_i^\top \mu_A = \begin{cases} \mu_A & \text{for subgroup 1,} \\ \mu_A + \nu_{A2} & \text{for subgroup 2,} \end{cases} \quad z_i^\top \mu_B = \begin{cases} \mu_B & \text{for subgroup 1,} \\ \mu_B + \nu_{B2} & \text{for subgroup 2.} \end{cases}$$

```
data("anticcp")
reitsmasub <- fitReitsmaSubgroup(data=anticcp,
                                TP=TP,
                                FP=FP,
                                FN=FN,
                                TN=TN,
                                study=study,
                                subgroup=generation)

reitsmasub
#>
#> Reitsma Subgroup Model
#> -----
#>
#> Number of studies      : 37
#> Number of subgroups   : 2
#> Model fit              : Converged
#> -2 log likelihood      : 533.37
#> AIC                    : 547.37
#> BIC                    : 558.646
#>
#>
#> Use summary() for parameter estimates.
summary(reitsmasub)
#> $estimates
```

```

#>               Estimate Std_Error
#> mu_A.CCP1      -0.09654001 0.22032136
#> mu_B.CCP1       3.44670662 0.29824593
#> mu_A.CCP2       0.86603422 0.12087713
#> mu_B.CCP2       3.01649071 0.16223893
#> sigma2_A.sens   0.35983136 0.10217750
#> sigma2_B.spec   0.53992184 0.18016162
#> sigma_AB        -0.19685042 0.09835584
#> nu_A.CCP2       0.96255893 0.25134200
#> nu_B.CCP2      -0.43020026 0.33771277
#>
#> $sensspec
#>      type      Orig conflevel  CI_Lower  CI_Upper
#> mu_A.CCP1 sens 0.4758837      0.95 0.3708990 0.5830440
#> mu_B.CCP1 spec 0.9691328      0.95 0.9459436 0.9825577
#> mu_A.CCP2 sens 0.7039198      0.95 0.6522898 0.7508123
#> mu_B.CCP2 spec 0.9533136      0.95 0.9369386 0.9655927
#>
#> $RutterGatsonis_recovered
#>      Lambda      Theta      beta sigma2_alpha sigma2_theta
#> CCP1 4.121053 1.278040 0.2028944 0.4878451 0.3188117
#> CCP2 3.727490 1.950972 0.2028944 0.4878451 0.3188117

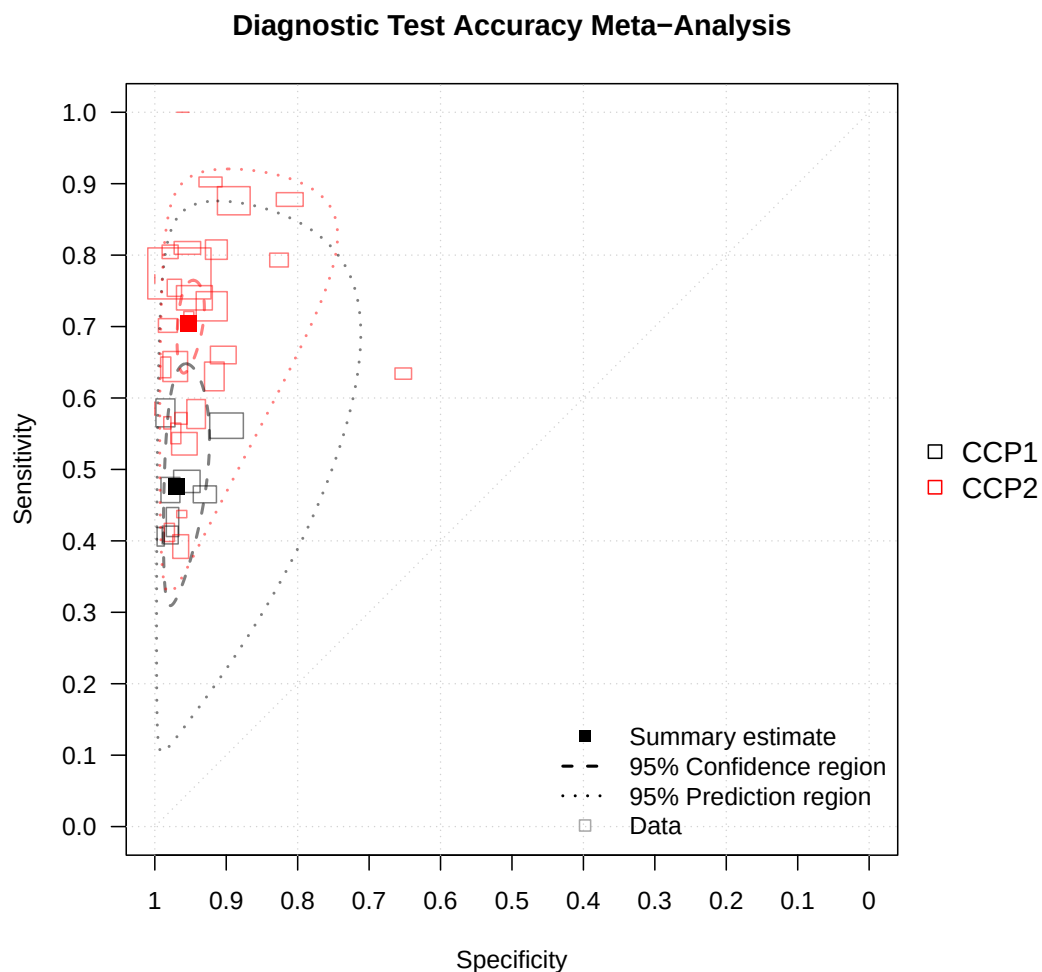
```

How do I get a summary plot of the Reitsma model?

```

plot(reitsmasub,
     scale=0.01,
     nudge_legend=-0.2,
     size="se",
     col=c("black","red"))

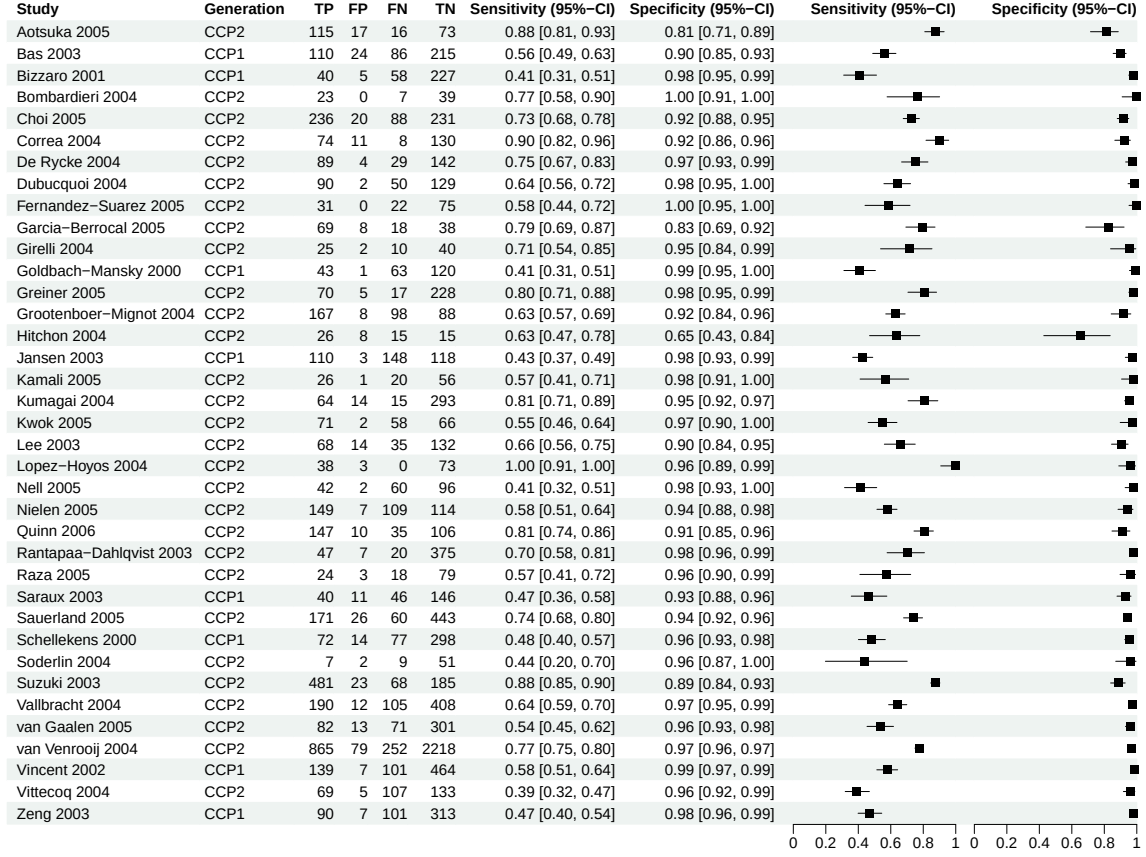
```



How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(reitsmasub, subgroup_label="Generation")
```



How do we include covariates in the Rutter and Gatsonis model?

The number of diseased individuals from study i who test positive is denoted by $y_{i1} \sim \mathcal{B}(n_{i1}, \pi_{i1})$.

Similarly, the number of non-diseased individuals who test positive is $y_{i2} \sim \mathcal{B}(n_{i2}, \pi_{i2})$.

Now we introduce a p -dimensional design-vector z_i including study level covariates. Consequently, at the study level, we have

$$\text{logit}(\pi_{ij}) = (z_i^\top \theta_i + z_i^\top \alpha_i x_{ij}) \exp(-z_i^\top \beta x_{ij}),$$

$$z_i^\top \alpha_i \sim \mathcal{N}(z_i^\top \Lambda, \sigma_\alpha^2), \quad z_i^\top \theta_i \sim \mathcal{N}(z_i^\top \Theta, \sigma_\theta^2),$$

where

$$x_{ij} = \begin{cases} -0.5 & \text{for non-diseased individuals,} \\ 0.5 & \text{for diseased individuals.} \end{cases}$$

Let's assume that z_i includes a single categorical covariate representing three subgroups. Then using dummy coding with subgroup 1 being the reference, we have

$$\begin{aligned} z_i^\top \theta_i &= \begin{cases} \theta_i & \text{for subgroup 1,} \\ \theta_i + \gamma_2 & \text{for subgroup 2,} \\ \theta_i + \gamma_3 & \text{for subgroup 3,} \end{cases} & z_i^\top \alpha_i &= \begin{cases} \alpha_i & \text{for subgroup 1,} \\ \alpha_i + \xi_2 & \text{for subgroup 2,} \\ \alpha_i + \xi_3 & \text{for subgroup 3,} \end{cases} \\ z_i^\top \beta &= \begin{cases} \beta & \text{for subgroup 1,} \\ \beta + \delta_2 & \text{for subgroup 2,} \\ \beta + \delta_3 & \text{for subgroup 3,} \end{cases} \\ z_i^\top \Lambda &= \begin{cases} \Lambda & \text{for subgroup 1,} \\ \Lambda + \xi_2 & \text{for subgroup 2,} \\ \Lambda + \xi_3 & \text{for subgroup 3,} \end{cases} & z_i^\top \Theta_i &= \begin{cases} \Theta & \text{for subgroup 1,} \\ \Theta + \gamma_2 & \text{for subgroup 2,} \\ \Theta + \gamma_3 & \text{for subgroup 3.} \end{cases} \end{aligned}$$

Notes:

- By default, `fitRutterGatsonisSubgroup()` assumes a common shape across all subgroups, which implies $\delta_2 = \delta_3 = 0$ for the case above.
- Moreover, for prediction `fitRutterGatsonisSubgroup()` uses the prediction matrix

$$Z_{\text{pred}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

for the case above and therefore recovers the threshold, accuracy, and shape parameters for each subgroup as if it were the reference group.

How do I perform a Rutter and Gatsonis meta-regression with a single categorical study-level covariate?

```
data("RF")
RF2      <- RF[RF$method %in% c("LA","ELISA","Nephelometry"),]
RF2$method <- factor(RF2$method,levels=c("LA","ELISA","Nephelometry"))
ruttergatsonissub <- fitRutterGatsonisSubgroup(data=RF2,
```

```

TP=TP,
FP=FP,
FN=FN,
TN=TN,
study=study,
subgroup=method)

```

```

ruttergatsonissub
#>
#> Rutter & Gatsonis Subgroup Model
#> -----
#>
#> Number of studies      : 47
#> Number of subgroups   : 3
#> Model fit              : Converged
#> -2 log likelihood     : 753.722
#> AIC                    : 771.722
#> BIC                    : 788.374
#>
#>
#> Use summary() for parameter estimates.
summary(ruttergatsonissub)
#> $estimates
#>
#>               Estimate Std. Error
#> Lambda_LA      2.45630704 0.32357707
#> xi_ELISA       0.24853054 0.43954126
#> xi_Nephelometry 0.33141751 0.44260069
#> Theta_LA      -0.55005213 0.21334057
#> gamma_ELISA    -0.19625909 0.26096239
#> gamma_Nephelometry 0.49586320 0.26223094
#> beta_LA        0.19768314 0.16983954
#> delta_ELISA    0.00000000 0.00000000
#> delta_Nephelometry 0.00000000 0.00000000
#> sigma2_alpha   1.27791582 0.30852514
#> sigma2_theta   0.47684216 0.11344574
#> Lambda_LA      2.45630704 0.32357707
#> Lambda_ELISA   2.70483758 0.32695434
#> Lambda_Nephelometry 2.78772454 0.30577914
#> Theta_LA      -0.55005213 0.21334057
#> Theta_ELISA    -0.74631122 0.20991330
#> Theta_Nephelometry -0.05418893 0.21213149
#> beta_LA        0.19768314 0.16983954

```

```

#> beta_ELISA          0.19768314 0.16983954
#> beta_Nephelometry   0.19768314 0.16983954
#> logitsens          0.82616754 0.28629967
#> logitsens          1.05130793 0.28606921
#> logitsens          1.12639409 0.27689507
#> sens               0.69554397 0.06062755
#> sens               0.74102598 0.05489854
#> sens               0.75517283 0.05119425
#>
#> $sensspec
#>      subgroup      spec conflevel logitsens Std_Error CI_Lower CI_Upper
#> 1      LA 0.8461538      0.95 0.8261675 0.2862997 0.2650305 1.387305
#> 2      ELISA 0.8461538      0.95 1.0513079 0.2860692 0.4906226 1.611993
#> 3 Nephelometry 0.8461538      0.95 1.1263941 0.2768951 0.5836897 1.669098
#>      Sens SensCI_Lower SensCI_Upper
#> 1 0.6955440      0.5658725      0.8001616
#> 2 0.7410260      0.6202531      0.8336879
#> 3 0.7551728      0.6419160      0.8414556
#>
#> $Reitsma_recovered
#>      mu_A.sens mu_B.spec sigma2_A.sens sigma2_B.spec      sigma_AB
#> LA      0.6142827 1.962946      0.6534849      0.970378 -0.1573632
#> ELISA    0.5490645 2.316770      0.6534849      0.970378 -0.1573632
#> Nephelometry 1.2135916 1.598491      0.6534849      0.970378 -0.1573632
#>
#> $subgroups
#> [1] "LA"          "ELISA"         "Nephelometry"

```

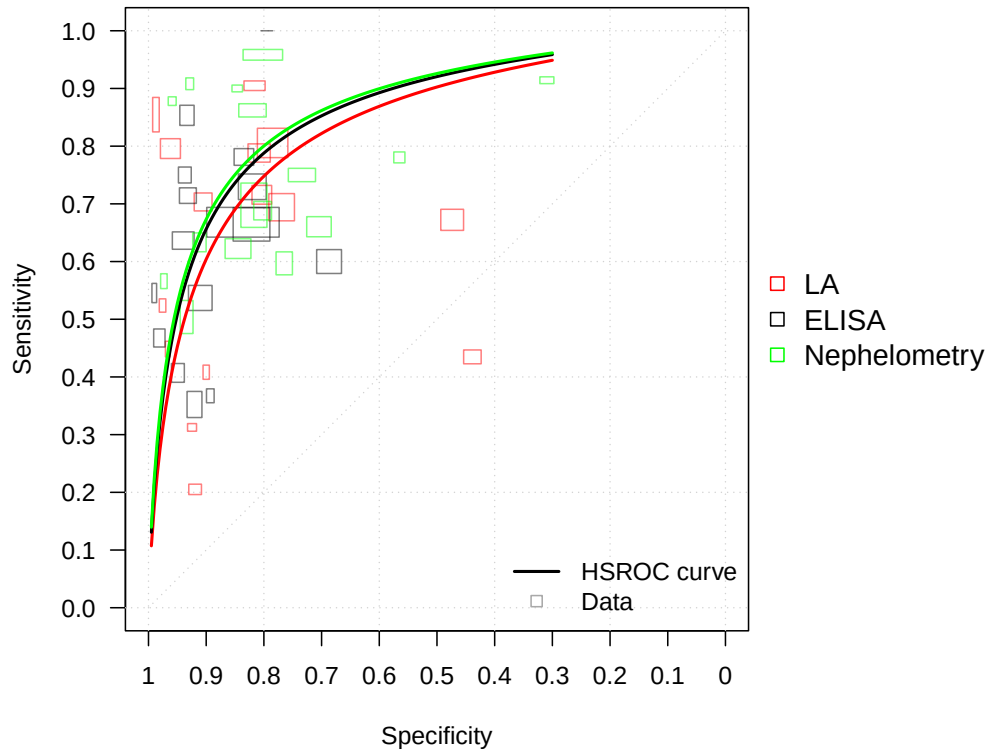
How do I get a summary plot of the Rutter and Gatsonis subgroup model?

```

plot(ruttergatsonissub,
     specrange=c(0.3,0.995),
     size="se",
     col=c("red","black","green"),
     scale=0.015)

```

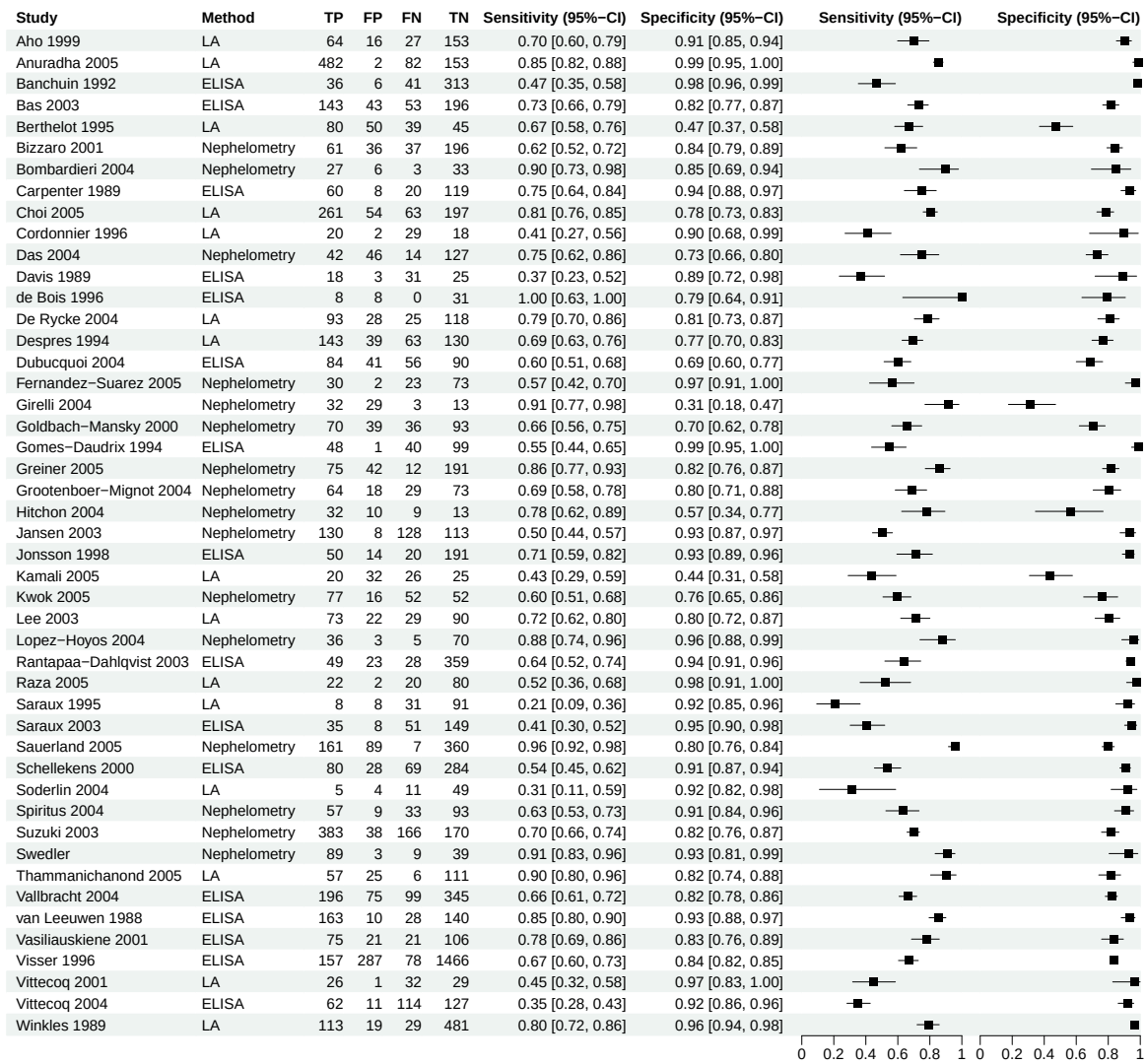

Diagnostic Test Accuracy Meta-Analysis



How do I get a coupled forest plot?

Note: Rendering forest plots may take longer in an interactive R session due to the underlying grid graphics. Performance is typically faster when knitting the vignette (e.g. via RMarkdown or Quarto).

```
forest(ruttergatsonissub, subgroup_label = "Method")
```



How do I use the general Rutter and Gatsonis regression function?

This method is for advanced users who feel comfortable specifying their own design and prediction matrices. For study-level covariates, the design matrix Z needs two identical consecutive rows per study, one for the diseased and one for the non-diseased. Of note, there are neither `plot()` nor `forest()` methods for `fitRutterGatsonisReg()`.

Let's reproduce the subgroup-analysis from before.

```

# Specify design matrix Z
Z <- model.matrix(~method,data=RF2)
# For study level-covariates, we need to two identical consecutive
# rows per study (diseased and non-diseased).
Z2 <- Z[rep(seq_len(nrow(Z)), each = 2), , drop = FALSE]
# Specify prediction matrix Z_pred
Z_pred <- matrix(c(1,0,0,1,1,0,1,0,1),ncol=3,nrow=3,byrow=T)

ruttermgatsonisreg <- fitRutterGatsonisReg(data=RF2,
                                           TP=TP,
                                           FP=FP,
                                           FN=FN,
                                           TN=TN,
                                           study=study,
                                           Z=Z2,
                                           Z_pred=Z_pred)

ruttermgatsonisreg
#>
#> Rutter & Gatsonis Regression Model
#> -----
#>
#> Number of studies : 47
#> Model fit          : Converged
#> -2 log likelihood : 753.722
#> AIC                : 771.722
#> BIC                : 788.374
#>
#>
#> Use summary() for parameter estimates.
summary(ruttermgatsonisreg)
#> $estimates
#>
#>      Estimate Std. Error
#> accuracy_coef 2.45631156 0.32357673
#> accuracy_coef 0.24852208 0.43954066
#> accuracy_coef 0.33141584 0.44260014
#> threshold_coef -0.55005084 0.21334149
#> threshold_coef -0.19625009 0.26096393
#> threshold_coef 0.49585695 0.26223232
#> shape_coef    0.19768191 0.16983950
#> shape_coef    0.00000000 0.00000000
#> shape_coef    0.00000000 0.00000000
#> sigma2_alpha  1.27791200 0.30852400

```

```

#> sigma2_theta      0.47684805 0.11344753
#> Lambda_Pred       2.45631156 0.32357673
#> Lambda_Pred       2.70483365 0.32695391
#> Lambda_Pred       2.78772741 0.30577879
#> Theta_Pred        -0.55005084 0.21334149
#> Theta_Pred        -0.74630093 0.20991415
#> Theta_Pred        -0.05419388 0.21213237
#> beta_Pred         0.19768191 0.16983950
#> beta_Pred         0.19768191 0.16983950
#> beta_Pred         0.19768191 0.16983950
#> logitsens         0.82617129 0.28629952
#> logitsens         1.05130416 0.28606887
#> logitsens         1.12639652 0.27689490
#> sens              0.69554476 0.06062743
#> sens              0.74102525 0.05489857
#> sens              0.75517328 0.05119416
#>
#> $sensspec
#>      spec conflevel logitsens Std_Error  CI_Lower CI_Upper      Sens
#> 1 0.8461538      0.95 0.8261713 0.2862995 0.2650345 1.387308 0.6955448
#> 2 0.8461538      0.95 1.0513042 0.2860689 0.4906195 1.611989 0.7410253
#> 3 0.8461538      0.95 1.1263965 0.2768949 0.5836925 1.669101 0.7551733
#>   SensCI_Lower SensCI_Upper
#> 1   0.5658735   0.8001621
#> 2   0.6202524   0.8336873
#> 3   0.6419166   0.8414559

```

References

- Reitsma, J. B., et al. (2005). Bivariate analysis of sensitivity and specificity produces informative summary measures in diagnostic reviews. *Journal of Clinical Epidemiology*, 58(10), 982–990.
- Rutter, C. M., & Gatsonis, C. A. (2001). A hierarchical regression approach to meta-analysis of diagnostic test accuracy evaluations. *Statistics in Medicine*, 20(19), 2865–2884.
- Harbord, R. M., Deeks, J. J., Egger, M., Whiting, P., & Sterne, J. A. C. (2007). A unification of models for meta-analysis of diagnostic accuracy studies. *Biostatistics*, 8(2), 239–251.
- Riley, R. D., Ensor, J., Jackson, D., & Burke, D. L. (2018). Deriving percentage study weights in multi-parameter meta-analysis models. *Statistical Methods in Medical Research*, 27(10), 2885–2905.

Hoyer, A., Hirt, S., Kuss, O. (2018). Meta-analysis of full ROC curves using bivariate time-to-event models for interval-censored data. *Research Synthesis Methods*, 9(1), 62-72.

Deeks, J. J., Bossuyt, P. M., Leeflang, M. M., & Takwoingi, Y. (editors) (2023). *Cochrane Handbook for Systematic Reviews of Diagnostic Test Accuracy*. Version 2.0 (updated July 2023). Cochrane.